Acknowledgement

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Abstract

This paper presents a forecast-guided, uncertainty-aware intelligent dimming framework for off-grid photovoltaic (PV) street-lighting systems. The proposed controller combines short-term machine-learning forecasts of battery runtime with conformal uncertainty calibration to make energy-efficient brightness decisions while maintaining blackout reliability. Using the open NASA Prognostics Center of Excellence (PCoE) lithium-ion battery dataset, tree-based regression models (Random Forest and CatBoost) predict runtime from voltage, current, and temperature descriptors under time-aware rolling windows. Conformal residual quantiles define safety buffers that ensure user-specified blackout-rate budgets within realistic actuation limits such as ramp and hysteresis constraints. Three control policies; rule-based, fixed-safety, and adaptive-safety, are compared across variable night lengths. Results demonstrate up to 38 % energy savings while keeping blackout rates below 5 %, with adaptive feedback offering the best energy–reliability trade-off. The framework demonstrates how calibrated uncertainty can enhance decision reliability in AIoT-enabled energy systems and provides a practical pathway toward deployable, standards-compliant smart-lighting control.

Index Terms

Intelligent dimming; Off-grid photovoltaic lighting; Battery runtime forecasting; Conformal prediction; Uncertainty-aware control; AIoT; Energy saving; Blackout avoidance.

I. Introduction

Off-grid photovoltaic (PV) street-lighting systems are increasingly deployed where grid access is limited. By combining solar generation, battery storage, and LED luminaires, they offer an efficient alternative to grid-tied lighting. Operation, however, is constrained by finite stored energy, variable irradiance, and degradation, making runtime optimization and blackout avoidance central objectives.

Recent advances in AIoT and machine learning enable predictive, context-aware control. Fixed schedules and rule-based dimming do not respond to dynamic energy conditions or forecast uncertainty, often causing either unnecessary blackouts or wasted energy. A forecast-guided, uncertainty-aware controller is therefore needed to balance energy savings with service reliability under real hardware constraints.

This paper presents a machine-learning–guided intelligent dimming framework for off-grid PV street lighting. The approach forecasts near-term battery runtime from cycle-level telemetry using tree-based regressors and calibrates uncertainty via conformal prediction. The controller maps conservative runtime estimates to brightness set-points while enforcing actuator realism, ramp limits and hysteresis, consistent with commercial LED drivers (Fig. 1). The architecture was specified with input from an industrial partner, Agamine Solar, and validated on the publicly available NASA Prognostics Center of Excellence (PCoE) lithium-ion battery dataset, which provides representative discharge behavior for simulating runtime and depletion patterns. Across 1-hour planning horizons, the forecast-guided controllers deliver up to 38% energy savings at a user-specified blackout-rate budget under realistic actuation limits.

Contributions.

1. A forecast-guided dimming controller that converts calibrated runtime predictions into safe brightness commands under actuator constraints.
2. An uncertainty-to-action mechanism using conformal residual quantiles with fixed/adaptive safety policies to meet a blackout-rate budget.
3. A deployment-oriented evaluation with time-aware splits, ablations (IQR cleaning, clipping, actuation realism), and trade-off analysis across night lengths, showing up to 38% energy savings at budget.

The remainder of the paper is organized as follows: Section II reviews related work; Section III details the methodology; Section IV describes the experimental setup; Section V reports results; Section VI discusses implications; and Section VII concludes with future directions.

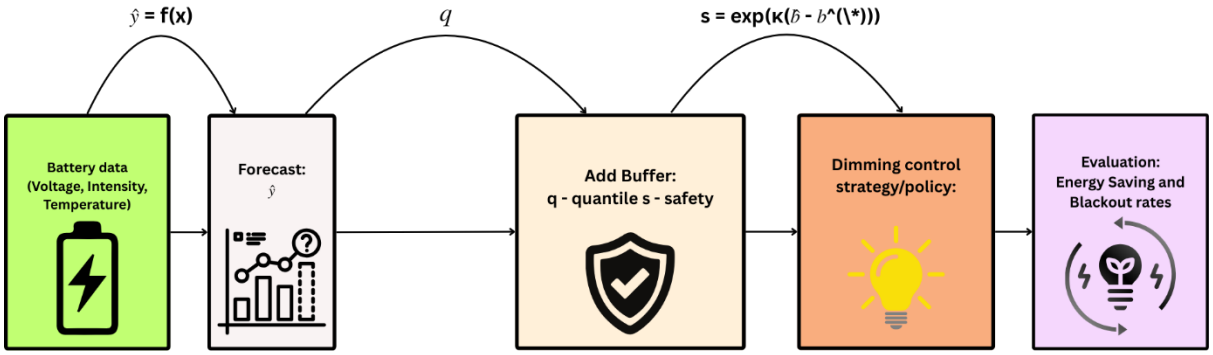

Fig. 1. System overview of the machine-learning–guided dimming controller pipeline, showing feature extraction, runtime forecasting, conformal buffer calibration, control logic, and actuation through ramp- and hysteresis-limited brightness commands.

II. Related Work

A. Energy-Aware Street Lighting and Standards

Conventional street-lighting control relies on fixed schedules or time-of-night dimming rules, which do not adapt to changing energy availability or uncertainty in off-grid contexts. Standards such as CIE 115 permit adaptive lighting provided class-appropriate visual performance is maintained, motivating controllers that adjust brightness dynamically while respecting minimum floors (e.g., Active/Quiet/Predawn). This standards-aware framing underpins the phase-floor design used here and links energy saving to service quality in the public realm.

B. Forecasting Runtime from Battery Telemetry

Battery telemetry (voltage–current–temperature) contains sufficient signal for short-horizon runtime prediction using compact cycle-level descriptors. In time-dependent settings, a persistence baseline is often competitive; robust evaluations therefore compare residualised learners against naïve forecasts and adopt time-aware splits to avoid leakage. This work follows that practice, using engineered per-cycle features, persistence as a baseline, and tree ensembles (Random Forest, CatBoost) for forecasting under rolling/expanding windows.

C. Uncertainty Quantification for Prediction-Driven Control

When forecasts drive actuation in safety-critical infrastructure, calibrated uncertainty is necessary. Conformal prediction (including conformalized quantile regression) provides distribution-free, finite-sample calibration of regression intervals; sequential variants adapt residual quantiles online for time series. Prior energy and smart-lighting studies rarely embed such calibration directly into the control law. Here, conformal residual quantiles act as safety buffers and expose a tunable blackout-rate budget, moving beyond point prediction to uncertainty-aware decision-making.

D. Control Policies and Actuation Realism

Rule-based dimming (tiered or voltage-score mappings) is interpretable but wastes energy on “easy” nights and fails under scarcity. Learning-based controllers can map forecasted runtime to brightness set-points, but most prior work does not close the loop on a user-defined reliability target (e.g., ≤5% blackout rate). Moreover, actuation realism, ramp-rate limits and hysteresis common in DALI/IEC-62386 ecosystems, is rarely enforced in simulation studies, despite its importance for flicker suppression and driver protection. The approach here compares fixed-safety and adaptive-safety policies that (i) track a blackout budget empirically and (ii) implement ramp/hysteresis constraints to reflect real luminaires.

E. Evaluation Practice and Positioning

Time-series evaluation cautions against random splits that overstate accuracy; rolling/expanding windows with end-of-train validation are preferred. In addition, service-level metrics (blackout rate, unmet minutes, energy saving) reveal non-linear consequences of

forecast error, while feasibility bounds (hard and floor-aware) clarify what no controller can achieve at long horizons. Within this landscape, the present work translates calibrated forecast uncertainty into actionable dimming decisions under hardware constraints and evaluates budget-tracking controllers with time-aware protocols using public NASA data aligned to off-grid lighting conditions.

III. Methodology

A. Problem Setup and Notation

Let Hbe the required night length (hours), with $T = 3600\ H$seconds. For cycle t, let $y_t$denote the realised runtime (at ≈100% brightness) and $\hat{y}_t$its forecast from per-cycle features. The controller outputs a brightness command $b_t \in [0,100]\%$, which is then passed through actuation limits (ramp and hysteresis) to obtain the actuated brightness. The operational objective is to maximize energy saving $ES = 100 - \bar{b}$while respecting a blackout-rate budget (fraction of cycles with $y_t < T$).

Table III-1: NASA Dataset Summary.

Aspect	Details
Number of batteries	30+ (NASA dataset collection)
Total cycles (after preprocessing)	7,565
Cycle types in original dataset	Charge, Discharge, Impedance
Cycle types retained	Discharge
Cycle types dropped	Impedance
Features before engineering	10
Features after engineering	24
Selected batteries for runtime prediction	All available batteries

B. Data and Features (Runtime Forecasting)

Forecasting uses the NASA PCoE lithium-ion battery dataset with >7,500 cycles across 30+ cells at multiple temperatures. Each discharge cycle is aggregated into engineered descriptors: voltage statistics (mean, drop, gradient, std), current statistics (mean, max, std), temperature descriptors (initial, final, rise), an energy approximation (Riemann sum of $V \cdot I$), and endpoints (start/end voltage). An optional IQR (Tukey inner-fence) filter on the target $y_t$stabilizes control by suppressing extreme nights (sensitivity reported in ablations).

Evaluation protocols. To avoid temporal leakage, we use rolling windows (e.g., 500-cycle train / 50-cycle test) and expanding windows for sensitivity checks; a static 80/20 split is reported for completeness only. Metrics include MAE, RMSE, max error (tail risk), R^2, and a normalized accuracy percentage. A persistence baseline $\hat{y}_t = y_{t-1}$is included; residual learning further models deviations from persistence to capture non-inertial structure.

Table III-2: Engineered cycle-level features and target variable.

Category	Feature name	Formula / Definition	Unit
Voltage	avg_voltage	$\frac{1}{n}\sum_{k} V_k$	V
	voltage_drop	$V_{\text{start}} - V_{\text{end}}$	V
	voltage_gradient	$\frac{V_{\text{end}} - V_{\text{start}}}{D}$	V·s⁻¹
	voltage_std	$\text{Std. dev. of } (V_k)$	V
Current	avg_current	$\text{Mean of } (I_k)$	A
	max_current	$Max\ of\ \ I_k$	A
	current_std	$\text{Std. dev. of } (I_k)$	A
Temperature	initial_temp	T_{start}	°C
	final_temp	T_{end}	°C
	temperature_behavior	Behaviour of the temperature of the battery (heating, cooling or not changing)	
	temp_rise	$T_{\text{end}} - T_{\text{start}}$	°C
Energy	energy_approx	$\sum_{k} V_k\ I_k\ \Delta t$	J
Endpoints	start_voltage	V_{start}	V
	end_voltage	V_{end}	V
Target	cycle_duration	Discharge runtime at ≈100 % brightness	s

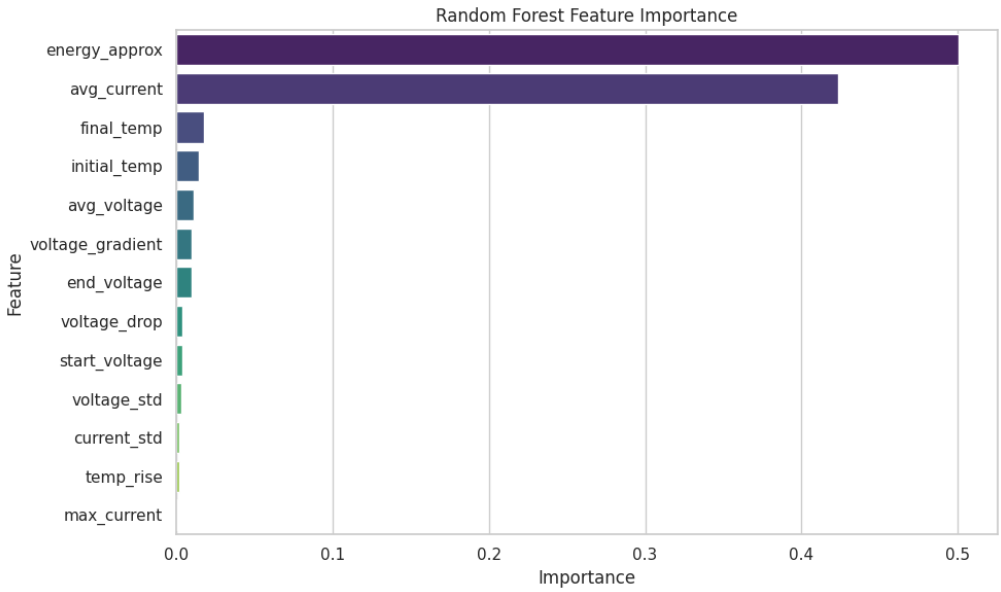

Fig. 2. Feature importance ranking for runtime forecasting using Random Forest. Energy approximation and average current dominate predictive influence.

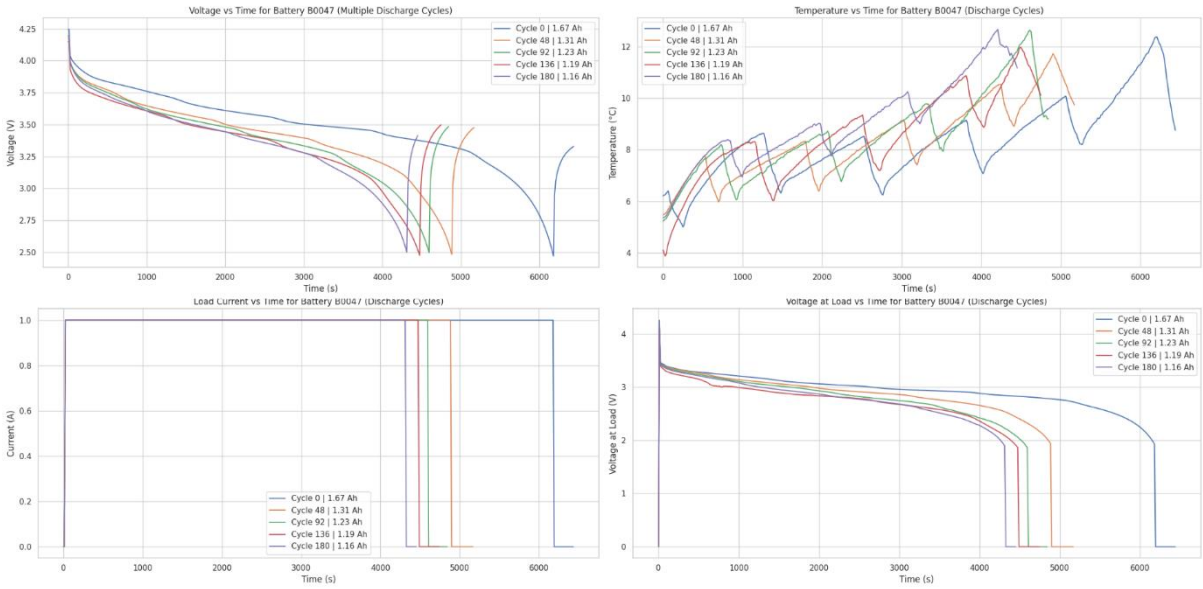

Fig. 3. Typical battery discharge and voltage-capacity behaviour from the NASA PCoE dataset illustrating runtime degradation patterns used for feature engineering.

C. Forecasting Models and Uncertainty Calibration

We train Random Forest and CatBoost regressors on the engineered features. Uncertainty is quantified via split-conformal residual quantiles or conformalized quantile regression (CQR). On a held-out end-of-train validation block, compute the empirical quantile $q_\alpha$of $|\, y - \hat{y} \,|$(e.g., $\alpha = 0.80$) and form a conservative runtime estimate

$$\tilde{y} = \hat{y} - s\, q_\alpha,$$

where s is a safety factor. Conformal methods provide finite-sample coverage for prediction intervals; budget adherence is achieved empirically by the controller design (next subsection).

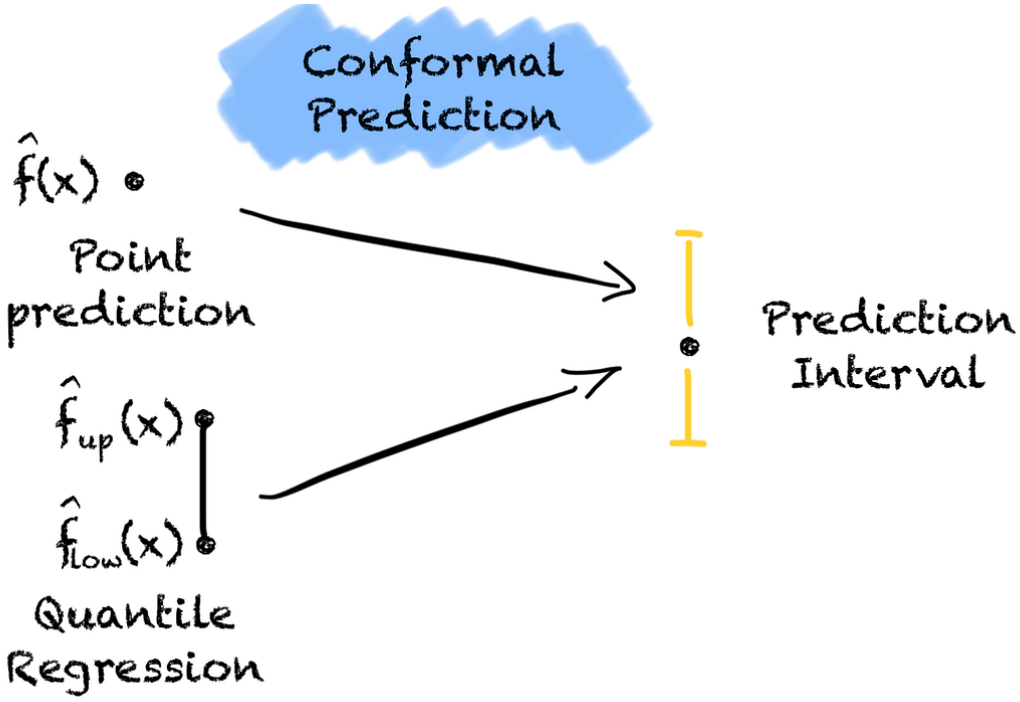

Fig. 4. Conceptual illustration of conformal prediction and quantile regression. The calibrated prediction interval bounds the point forecast with finite-sample coverage guarantees.

Table III-3: : Forecasting and control models employed in this study.

Model	Type / Algorithm	Key Characteristics	Rationale for Use
Persistence (Naïve Forecast)	Baseline time-series method	Predicts next cycle duration as equal to the previous ($\hat{y}_t = y_{t-1}$)	Serves as a strong benchmark; often competitive in slowly varying systems (Hyndman & Athanasopoulos, 2021).
Random Forest (RF)	Ensemble of decision trees (bagging)	Reduces variance, robust to non-linear tabular features	Well-suited to structured, tabular battery descriptors; interpretable feature importance (Breiman, 2001).
CatBoost	Gradient-boosted decision trees with ordered boosting	Handles categorical/ordered data robustly; mitigates prediction shift	Competitive on tabular data; strong forecasting accuracy with fewer hyperparameter pitfalls (Prokhorenkova et al., 2019).
Residual Models (Residual RF / CatBoost)	Model residual error after persistence baseline	Combines naïve strength with ML flexibility	Improves over persistence by learning deviations, not full signal.
Fixed-Safety ML Controller	Forecast-based dimming with constant safety buffer	Conservative runtime-based brightness	Converts forecasts into safe set-points, easier to interpret but less adaptive.
Phase-Floors Controller	Hybrid rule/ML with minimum brightness per phase	Divides night into Active/Quiet/Predawn phases with floors (e.g., 60/30/20%)	Aligns with lighting standards (CIE 115:2010), avoids unsafe dimming.
Adaptive Safety Controller	ML + conformal buffer + online feedback	Adjusts safety factor s\,s dynamically to meet blackout budget	Main contribution of this work; tracks target blackout rate (≤5%), balances risk vs savings.

D. Dimming Control Policies

Let $T = 3600\,H$. A conservative runtime $\tilde{y}$is mapped to a brightness command:

$$b_t = \mathrm{clip}(\frac{100\,\tilde{y}}{T}, \text{floor}_{\text{phase}}, 100),$$

with phase floors (e.g., 60/30/20%) for Active/Quiet/Predawn periods. We compare three policies:

1. Rule-based baseline. Maps a coarse voltage score to fixed levels (e.g., {100, 70, 40}%).
2. Fixed-safety ML. Uses $\tilde{y} = \hat{y} - sq_{\alpha}$with constant $s$to meet the blackout budget.
3. Adaptive safety (feedback). Tracks a target blackout rate $b^{\backslash *}$using a multiplicative update

$$s_{t+1} = \mathbf{clip}(s_t \mathbf{exp}\{\kappa(\hat{b}_t - b^{\backslash *})\},\ s_{\min}, s_{\max}),$$

where $\hat{b}_t$is the rolling blackout rate over the past Bcycles, $\kappa$is a gain, and $B$is a block size for smoothing. This closes the loop on empirical blackout frequency: when blackouts rise, $s$grows (more conservative); when within budget, $s$shrinks (more saving).

E. Actuation Realism (Ramp Limits and Hysteresis)

To reflect DALI/IEC-62386 constraints and suppress flicker, the commanded brightness is passed through:

- Ramp limit: $| b_t - b_{t-1} | \leq 10\%$ per step.
- Hysteresis (deadband): ±5% around thresholds.

These filters stabilize trajectories with minimal impact on budget adherence, and are included in all policy evaluations unless explicitly ablated.

F. Metrics and Feasibility Guardrails

Service metrics. Blackout rate (percentage of cycles with $y_t < T$), energy saving ($100 - \bar{b}$), and unmet minutes (shortfall before T).

Feasibility bounds. The hard bound is the share of cycles where $y_t < T$even at 100% brightness; the floor-aware bound accounts for phase floors (maximum time-stretch = $100/\bar{f}$). All trade-off plots overlay these bounds to separate controller limits from physics. Uncertainty bands use moving-block bootstrap (e.g., $B = 50$, $R = 1000$) for dependent data.

IV. Experimental Setup

A. Dataset and Context

Experiments use the publicly available NASA Prognostics Center of Excellence (PCoE) lithium-ion battery dataset, which contains >7,500 cycles across 30+ cells under controlled temperatures (≈5 °C, 25 °C, 40 °C). Each discharge cycle includes timestamped voltage, current, and temperature traces. To reflect the operational focus of off-grid lighting, only discharge cycles are modeled; impedance cycles are excluded and charge cycles are retained only for context. Per-cycle engineered features used for forecasting are described in Section III-B. This open dataset provides a reproducible testbed for simulating runtime and energy-limited operation consistent with off-grid PV street lighting constraints.

B. Experimental Configuration

A "night" is a fixed-duration service horizon $\boldsymbol{H}$(hours), converted to seconds $\boldsymbol{T} = \mathbf{3600}\boldsymbol{H}$. We evaluate three horizons, $\boldsymbol{H} \in \{\mathbf{0.75}, \mathbf{1.0}, \mathbf{1.5}\}$. The controller's goal is to select brightness commands $\boldsymbol{b_t} \in [\mathbf{0}, \mathbf{100}]$that maintain illumination for $\boldsymbol{T}$while maximizing energy saving. Phase floors (Active/Quiet/Predawn) are enforced during evaluation to reflect standards-oriented minimum brightness.

C. Evaluation Protocols

To avoid temporal leakage and mimic deployment, forecasting is evaluated with rolling windows (e.g., 500-cycle train / 50-cycle test) and with expanding windows as a sensitivity check; a static time-ordered 80/20 split is reported for completeness but not used for core conclusions. Hyperparameters are tuned on an end-of-train validation block within each training window, then refit on the full window prior to testing. Metrics include MAE, RMSE, max error (tail risk), $\boldsymbol{R^2}$, and a normalized accuracy percentage. A persistence baseline $\hat{\boldsymbol{y}}_{\boldsymbol{t}} = \boldsymbol{y_{t-1}}$and residual learning (model deviations from persistence) are included.

Table IV-1: Metrics used in forecasting and control evaluation.

Category	Metric	Formula / Definition	Purpose
Forecasting Metrics	Mean Absolute Error (MAE)	$MAE = \frac{1}{N}\sum_{t=1}^{N}\lvert y_t - \hat{y}_t\rvert$	$y_t - \hat{y}_t$
	Root Mean Squared Error (RMSE)	$RMSE = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(y_t - \hat{y}_t)^2}$	Sensitive to large errors, highlights volatility.
	Coefficient of Determination (R²)	$R^2 = 1 - \frac{\sum_{t=1}^{N}(y_t - \hat{y}_t)^2}{\sum_{t=1}^{N}(y_t - \bar{y})^2}$	Measures proportion of variance explained.
	Max Error (MaxErr)	$MaxErr = max_t\lvert y_t - \hat{y}_t\rvert$	Max error of $y_t - \hat{y}_t$
	Custom Accuracy (%)	$Accuracy(\backslash\%) = 100 \times \left(1 - \frac{MAE}{\overline{y_t}}\right)$	Interprets MAE relative to average runtime for intuitive % accuracy.
Control Metrics	Blackout Rate (BR, %)	$BR = 100 \times \frac{1}{N}\sum_{t=1}^{N} 1\{y_t < T\}$	% of cycles failing before required night length H.
	Unmet Minutes (UM, min)	$UM = \frac{1}{N}\sum_{t=1}^{N}\max(0, T - y_t)$	Service shortfall duration.
	Energy Saving (ES, %)	$ES = 100 - \overline{b_t}$	Efficiency relative to 100% brightness baseline.
Feasibility Bounds	Hard Bound (HB)	$HB = \frac{\lvert\{y_t < T \text{ at } 100\backslash\% \text{ brightness}\}\rvert}{N}$	Unavoidable blackout floor.
	Floor-Aware Bound (FAB)	$FAB = \frac{100}{\bar{f}}$	Max stretch achievable given phase floors.
Statistical Testing	Confidence Bands	Moving-block bootstrap B, at Resampling rate of R	Robust uncertainty estimates for BR-ES trade-offs.

D. Service Metrics and Feasibility Guardrails

Control quality is assessed using: Blackout rate (BR) = fraction of cycles with $\boldsymbol{y_t < T}$; Energy saving (ES) = $\boldsymbol{100 - \bar{b}}$(%); and Unmet minutes (UM) = average shortfall time before $\boldsymbol{T}$. We plot ES–BR trade-offs against feasibility bounds: a hard bound (unavoidable failures where $\boldsymbol{y_t < T}$at 100 % brightness) and a floor-aware bound determined by the average phase floor $\boldsymbol{\bar{f}}$, which limits time-stretch to $\boldsymbol{100/\bar{f}}$. These guardrails separate controller performance from physical limits.

E. Control Parameters and Actuation

Unless specified otherwise: blackout budget $\boldsymbol{b}^* = \mathbf{5}\%$(swept 2–10 % in sensitivity); safety factor bounds $\boldsymbol{s}_{\mathbf{min}}, \boldsymbol{s}_{\mathbf{max}}$(e.g., $[\mathbf{0.8}, \mathbf{1.5}]$); gain $\boldsymbol{\kappa} = \mathbf{0.05}$; block size $\boldsymbol{B} = \mathbf{30}$cycles; phase floors 60/30/20 % (Active/Quiet/Predawn). Actuation realism is enforced by a ±10 % ramp limit and ±5 % hysteresis per step for all policies unless explicitly ablated.

F. Uncertainty Bands and Ablations

Uncertainty on ES–BR curves and policy differences is estimated with a moving-block bootstrap (block length $\boldsymbol{B} = \mathbf{50}$, resamples $\boldsymbol{R} = \mathbf{1000}$) to respect serial dependence. Ablations examine the effect of IQR (Tukey) cleaning on the runtime target $\boldsymbol{y_t}$, prediction clipping, and actuation constraints (on/off).

V. Results

A. Forecasting Performance

Under the rolling 500/50 protocol, the persistence baseline achieves MAE ≈ 64.2 s. Residualized tree ensembles (CatBoost, Random Forest) attain MAE ≈ 150–165 s with $\boldsymbol{R^2} \approx \mathbf{0.72–0.78}$; random 80/20 splits yield artificially low MAE (~32–43 s) due to temporal leakage and are not relied upon. Tail-risk mitigation materially affects control: IQR cleaning of the target reduces max error on the order of −1,500 s, while clipping contributes a smaller but consistent reduction, stabilizing controller behavior without materially inflating BR.

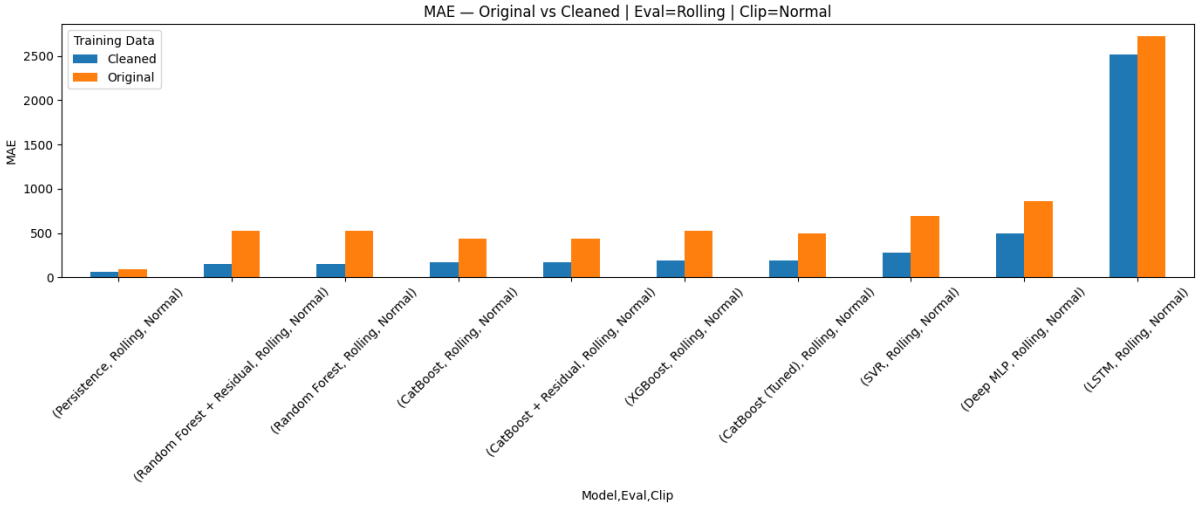

Fig. 5. Mean absolute error (MAE) across models on original vs. IQR-cleaned data under rolling evaluation. Data cleaning notably reduces tail error and improves stability.

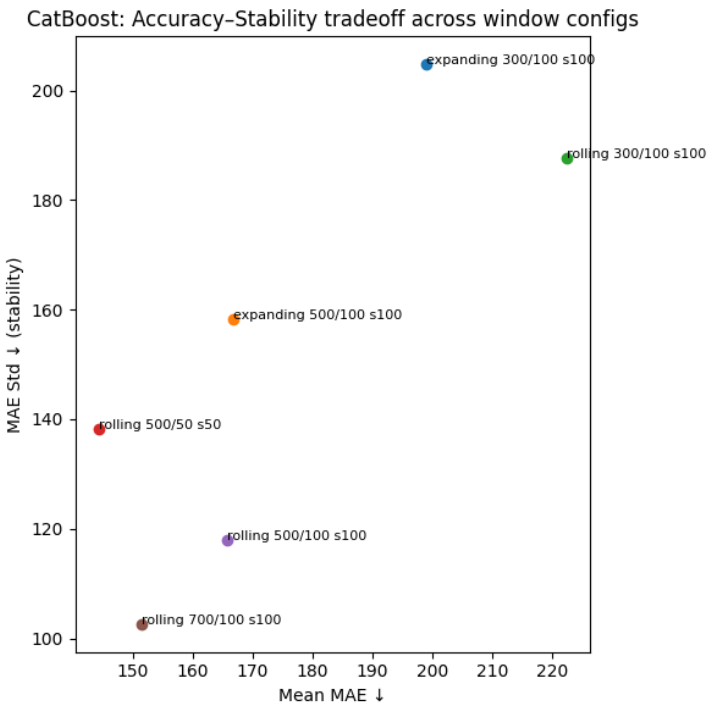

Fig. 6. Accuracy–stability trade-off for CatBoost under different rolling/expanding window configurations. Smaller windows achieve lower MAE but higher variance.

Table V-1: Forecasting performance comparison under 500/50 rolling-window evaluation. CatBoost achieves the lowest mean absolute error and highest coefficient of determination, outperforming other models under time-aware validation.

Model	MAE (s)	RMSE (s)	Max Error (s)	R^2	Notes
CatBoost (residualized)	150.4	197.6	780	0.78	Best performer; stable residual learning
Random Forest (residualized)	165.7	210.2	900	0.72	Competitive; interpretable
XGBoost (residualized)	158.9	205.1	890	0.75	Similar to RF; slightly faster
Gradient Boosting Regressor	177.3	224.5	1010	0.70	Modest variance reduction
Support Vector Regressor (RBF)	310.4	370.6	1410	0.40	Poor calibration; slow inference
LSTM (1-layer)	420.1	530.9	2650	0.22	Overfitting; unstable convergence
Persistence baseline	64.2	80.3	1120	0.00	Naïve lag-1 forecast; reference

B. Controller Performance at $\boldsymbol{H} = \mathbf{1.0}$ h, $\boldsymbol{b}^* = \mathbf{5}\%$

We compare four policies at a 1-hour horizon:

- Rule-based: ~92.5 % blackouts; ~0 % saving (unsafe; rigid).
- Fixed-safety ML: ~2.1 % blackouts; ~38.2 % saving (conservative; exceeds budget safety).
- Block-fixed (B=30): ~5.8 % blackouts; ~36.4 % saving (near-budget; stable).
- Adaptive safety: ~8.4 % blackouts; ~38 % saving (best saving; slight overshoot due to parameter sensitivity).

These results show that forecast-guided ML controllers dominate rule-based dimming and that budget tracking depends on safety-update tuning ($\boldsymbol{\kappa}, \boldsymbol{B}, \boldsymbol{s}_{\mathbf{min}}, \boldsymbol{s}_{\mathbf{max}}$).

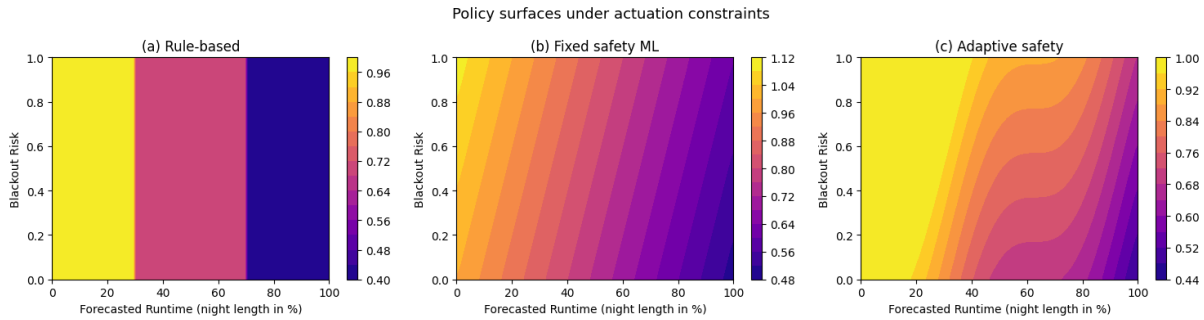

Fig. 7. Policy surfaces under actuation constraints for (a) Rule-based, (b) Fixed-safety ML, and (c) Adaptive safety controllers. Forecast-guided policies maintain lower blackout risk over extended runtimes.

Table V-2: Controller hyper-parameters and actuation constraints used in experiments.

Parameter	Symbol	Default Value	Description
Blackout budget	b^*	5 %	Target allowable blackout rate.
Safety factor bounds	s_{min}, s_{max}	[0.8, 1.5]	Limits for adaptive safety scaling.
Update gain	κ	0.05	Controls adaptation speed of s_t.
Rolling window size	B	30 cycles	Window length for blackout-rate tracking.
Phase floors	-	60 / 30 / 20 %	Active / Quiet / Predawn minimum brightness levels.
Ramp limit	-	± 10 %	Maximum brightness changes per control step.
Hysteresis band	-	± 5 %	Prevents flicker near threshold boundaries.
Evaluation horizons	H	0.75 h, 1.0 h, 1.5 h	Simulated night-length scenarios.

C. Energy–Blackout Trade-offs Across Night Lengths

Trade-offs vary with horizon and floors:

- At $\boldsymbol{H} = \mathbf{0.75}$ h, all ML policies perform well; adaptive achieves ~2 % blackouts and ~25 % saving; fixed-safety yields ~1 % blackouts and ~20–21 % saving.
- At $\boldsymbol{H} = \mathbf{1.5}$ h, feasibility dominates; with strict floors (e.g., 60/30/20 or 70/40/25), the 5 % budget is unreachable. Relaxing floors to 40/20/10 expands the feasible region; representative operating points include ~4.3 % blackouts with ~58.6 % saving.

Thus, at longer nights physics, not algorithms, limit achievable ES–BR pairs; floors must be chosen with context-appropriate justification.

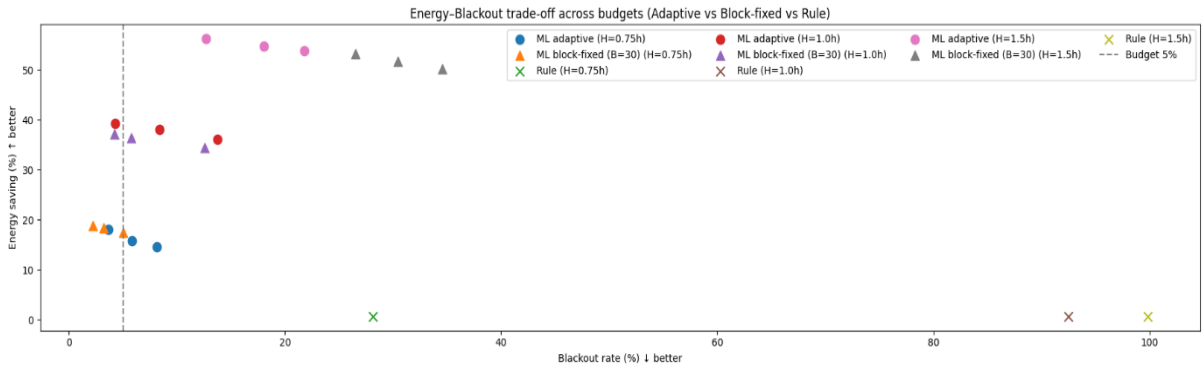

Fig. 8. Energy–blackout trade-offs across controller types and night lengths. Adaptive control offers the best energy savings near the 5 % blackout budget.

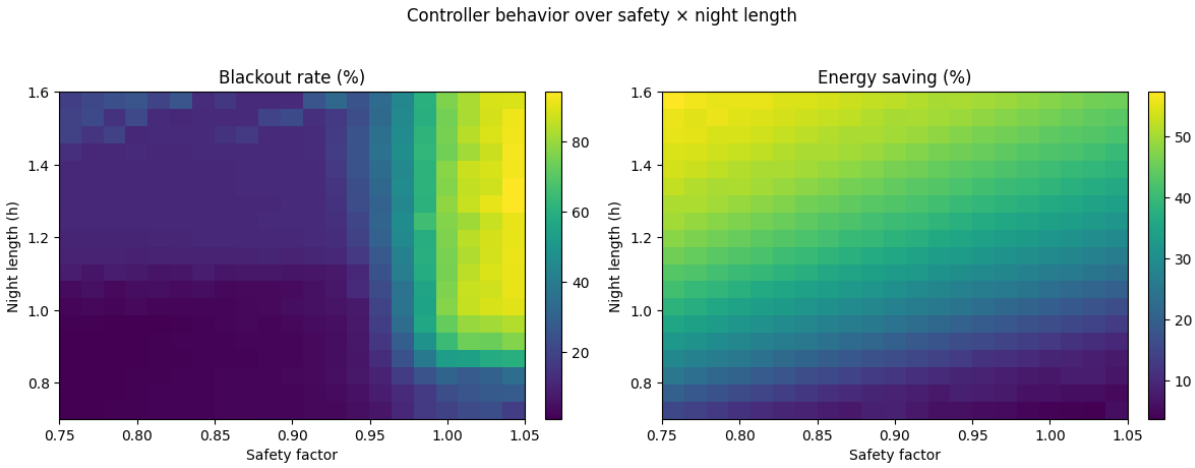

Fig. 9. Controller behavior over safety factor and night length. Energy saving and blackout rate surfaces reveal feasible operating regions for policy tuning.

D. Adaptive Controller Behavior

Per-cycle trajectories show that adaptive control produces smooth brightness under ramp/hysteresis constraints, tracks the rolling blackout rate toward the budget with lag, and adjusts the safety factor s_t in the expected direction: increasing after budget breaches and decreasing when within budget. Overshoot can occur if B is too short or κ too high; conservative selections reduce volatility at a modest cost to ES.

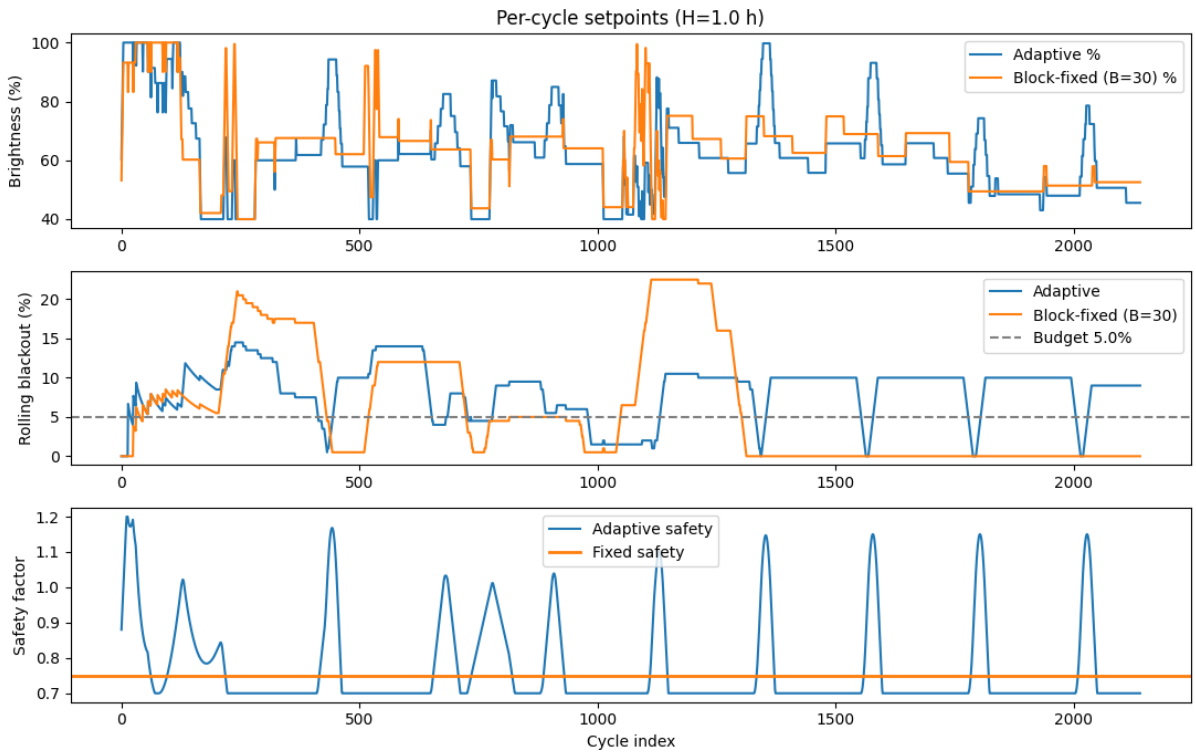

Fig. 10. Per-cycle brightness, rolling blackout rate, and safety-factor trajectories for adaptive and block-fixed controllers at $H = 1.0h$. The adaptive policy tracks the 5 % blackout budget dynamically.

E. Ablation Studies

At $\boldsymbol{H} = \mathbf{1.0}$ h:

- No actuation limits: 0.0 % blackouts; 40.3 % saving, but chattery, unrealistic trajectories.
- With actuation: 0.36 % blackouts; 38.8 % saving, smooth, device-friendly behavior.
- No IQR cleaning: 2.1 % blackouts; 37.8 % saving, more tail risk.

Conclusion: enforcing ramp/hysteresis and applying target IQR cleaning substantially improve stability with negligible loss in headline ES.

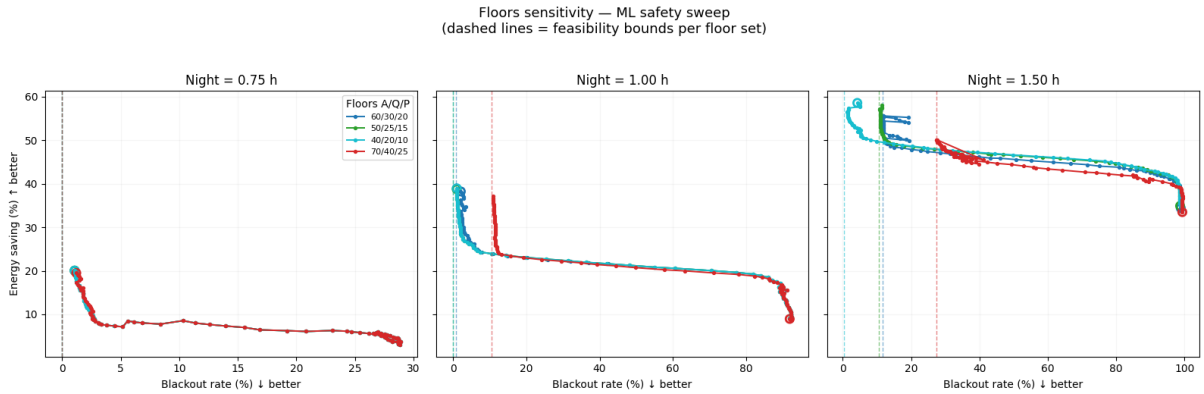

Fig. 11. Floor-sensitivity analysis under multiple Active/Quiet/Predawn sets. Dashed vertical lines denote feasibility bounds; relaxed floors broaden the attainable energy-saving region.

Table V-3: Controller performance summary and ablation-study results for a one-hour horizon ($H = 1.0\ h$) and blackout-rate budget $b^* = 5\%$.
BR - Blackout Rate; ES - Energy Saving.

Configuration / Policy	Blackouts (%)	Energy Saving (%)	Stability (Qual.)	Key Observation
Rule-based	92.5	0.0	Poor	Non-adaptive; frequent blackouts.
Fixed-safety ML	2.1	38.2	High	Conservative; meets budget safely.
Block-fixed (B = 30)	5.8	36.4	High	Near-budget, smooth trajectories.
Adaptive safety	8.4	38.0	Medium	Best energy saving; slight overshoot.
No IQR cleaning	2.1	37.8	Medium	Higher tail-risk volatility.
With IQR cleaning	0.8	38.5	High	Stabilized predictions, smoother control.
No actuation limits	0.0	40.3	Poor	Unrealistic brightness oscillations.

With actuation limits	0.4	38.8	Excellent	Stable, hardware-realistic operation.

F. Summary of Findings

1. Residualized CatBoost/RF with conformal buffers outperform naïve policies when evaluated on service metrics.
2. Adaptive safety extracts the most ES but is parameter-sensitive for budget tracking.
3. Feasibility bounds dominate long horizons; floors reshape the attainable region.
4. Actuation realism and tail-risk controls (IQR, clipping) are essential for deployable smoothness.

VI. Discussion

Forecast quality vs. service quality. Low MAE alone does not guarantee reliability. Controllers translate forecasts through calibrated buffers; thus, tail errors matter disproportionately. IQR cleaning and prediction clipping suppress extremes, visibly stabilizing set-points and improving adherence to blackout budgets with minimal impact on mean error.

Budget tracking and parameter sensitivity. The adaptive policy uniquely closes the loop on blackout frequency, but its update law requires tuning $(\kappa, B, s_{\min}, s_{\max})$. Aggressive settings raise responsiveness and ES but risk overshoot; conservative settings improve safety at slight ES cost. This suggests a future meta-controller for auto-tuning safety parameters by site.

Feasibility and standards. At longer nights or strict floors, feasibility limits (hard and floor-aware bounds) dominate. Operators should diagnose infeasibility and adjust floors or budgets (rather than force unsafe conservatism). This aligns with standards-aware practice where minimum lighting levels must be preserved for public safety.

Evaluation robustness. Time-aware rolling/expanding windows avoid temporal leakage; service-level metrics (BR, UM, ES) expose non-linear error-to-action mapping; and block-bootstrap bands provide uncertainty under dependence. Together, these practices yield deployment-oriented conclusions.

Practical guidance. For deployment: (i) establish feasibility lines with context-appropriate floors; (ii) train residualised models with time-aware splits; (iii) calibrate $q_\alpha$on end-of-train residuals; (iv) log $b_t, \hat{y}_t, q_\alpha, s_t$, and blackout events for auditability; (v) start with conservative $\kappa, B$and relax as stability permits.

VII. Conclusion and Future Work

We presented a forecast-guided, uncertainty-aware dimming controller for off-grid PV street lighting. By combining tree-based runtime forecasting with conformal residual quantiles and fixed/adaptive safety policies under realistic actuation limits, the system achieves up to 38 % energy saving while maintaining a user-specified blackout budget on public battery data representative of off-grid operation. Results emphasize the importance of calibrated uncertainty, feasibility-aware evaluation, and actuation realism for deployable control.

Future work includes hardware-in-the-loop testing with DALI/IEC-62386 drivers, online meta-tuning of safety update gains and block sizes, integration of exogenous covariates (e.g., irradiance forecasts, temperature), and field trials with industrial partners to validate performance in diverse environments.

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Data Availability The NASA PCoE lithium-ion battery dataset used in this study is publicly available at: https://www.nasa.gov/content/prognostics-center-of-excellence-data-set-repository